

Is the 360° street view redundant?

Phase-2 attention pruning — Qwen3.6-27B, $k=1$, $n=1663$ solved questions

The question

Each urban VQA question shows **4 street-view cameras** (fwd, bwd, cross_left, cross_right).

Hypothesis

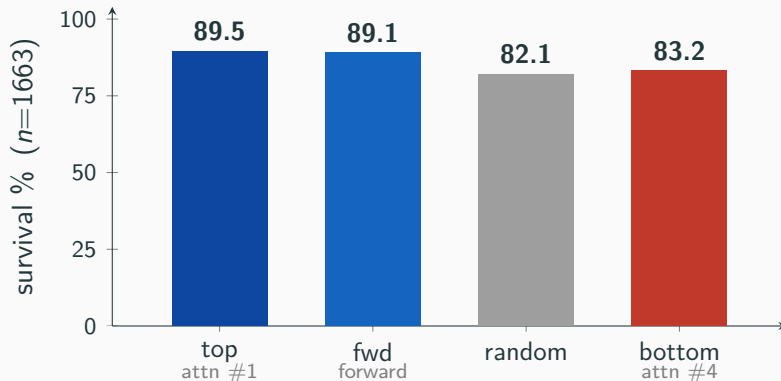
The full 360° view is **redundant**: the **one image the model attends to most** keeps the answer correct — and that image is the **forward camera**.

Method. Solve each question on all 4 images → keep only the single chosen image → re-ask greedily.

Survival % = fraction of solved questions still correct on the kept image.

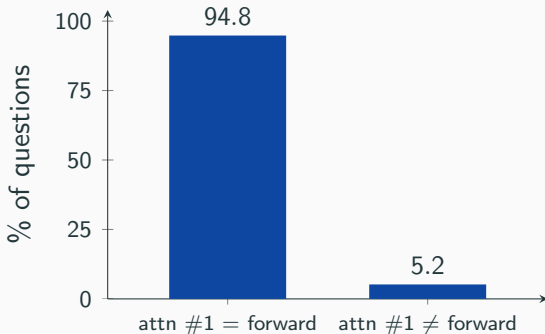
Arms: **top** (most-attended) | **fwd** (always forward) | random | **bottom** (least-attended).

One image is enough



top = **fwd** $\gg$ random. The single most-attended image keeps the answer correct **9 times in 10**.

The attended image is the forward camera



- Attention's #1 image is the forward camera in **94.8%** of questions.
- Pooled survival: **top** 89.5% $\approx$ **fwd** 89.1% — picking by attention and “always forward” land in the same place.
- Practical rule: **keep the forward image, drop the other three.**

Holds across every topic

topic	forward	random	gap
road surface	97.5	87.0	+10.5
junction type	91.2	89.0	+2.2
transit density	90.8	83.8	+7.0
road type	89.4	82.2	+7.2
urban density	86.9	79.8	+7.1
land use	87.1	79.8	+7.3
building height	85.8	76.9	+8.9
amenity richness	77.2	72.8	+4.4
pooled	89.1	82.1	+7.0

- ✓ The 360° view is **redundant** — **one** street-view image suffices (89.5% survival).
- ✓ That image is the **forward camera** (`along_fwd`), chosen by attention 94.8% of the time.
- ✓ The right image **beats a random one** by +7 pts, in **every** topic.

3× fewer image tokens per question, same answer.

Repro: `python attention_prune/analyze_prune_k1.py` | $n=1663$, sha b083f41.